

Towards a Neural ODE Generative Model of Cortical Dynamics

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Abstract—Cytoarchitecture studies have revealed that neurons in cortex are not arranged at random, but instead follow a stereotyped pattern - repeated throughout the cortex. This powerful canonical microcircuit (CMC) motif subtends learning, memory, and a vast array of complex cognitive behaviors. Our goal was to create a biologically plausible generative model, based on this circuitry, that could both perform complex tasks and generate realistic neural dynamics for cell populations. Towards this goal, we developed a neural ODE (nODE) implementation of the CMC model that translates this architecture into a learnable dynamical system - incorporating four distinct neural populations: spiny stellate cells, superficial pyramidal cells, inhibitory interneurons, and deep pyramidal cells. Each population is characterized by voltage and conductance states, resulting in an 8-dimensional system. We found that even a single CMC node achieved 96.77% accuracy on MNIST classification while maintaining biological plausibility through its recurrent excitatory and inhibitory dynamics. We further extend this architecture into hierarchical and recurrent configurations inspired by the organization of the human visual processing stream. The hierarchical model incorporated learnable inter-regional connectivity, while the recurrent version added bidirectional connections that mirror cortical processing. We found that increasing model depth and adding recurrent connections preserves interpretable dynamics while improving performance on more complex benchmark data sets. We suggest that this neural ODE implementation provides a computationally efficient framework that could serve as a generative model to guide neuromodulation and precision BCI systems.

Keywords—brain computer interface, BCI, brain computational model, neural ODE, canonical microcircuits

I. INTRODUCTION

Generative models of brain information processing represent a powerful tool for testing hypotheses of how neurons implement computations. To bridge Marr's levels, these generative models should be built with neurobiologically plausible subcomponents, reproduce brain activity data across scales, and be capable of performing complex cognitive tasks [1]. These brain computational models may lay the foundation for novel data generation, decoding, prediction, and interpretation schemes. Ideally, these models could be tuned to the structure and functional data of an individual brain to better inform brain-computer interfaces and precision neuromodulation therapy. Here we propose to leverage the canonical microcircuit model (CMC) developed in the computational neuroscience community [2], considered a fundamental module for computation in the brain. We will translate this model using the neural ordinary differential equation (nODE) approach, to scale this model for efficient training to accomplish classic vision tasks.

A. Canonical Microcircuits

Across the neocortex, a common stereotyped pattern of connectivity is found within microcolumns. These cortical micro columns (50 to 500 μm) are arranged orthogonal to the pial surface, containing approximately 10^2 to 10^4 neurons. Microcolumns typically contain all the major neural phenotypes found in the neocortex, each of which localizes to a specific layer of the cortex during ontogeny [3]. Techniques that combine cellular spike recordings, optogenetics, and histology have revealed that the columnar circuitry of the neocortex follows a common pattern characterized by recurrent excitatory and inhibitory feedback loops at the local level. This pattern is observed not only in sensory and motor areas, but also in association areas that subtend the highest level of cognitive processing [4]. The column has thus been viewed as a fundamental unit of information integration and functional specificity.

This columnar flow of information processing (up and down) between layers appears to subserve a vast complexity of cognitive processing due to 1) their interaction in large quantity, 2) their embedding within an organized small world network topology, 3) their asymmetric extrinsic connections, and 4) individual tuning properties. Differential weighting of layers in the cortex is crucial for working memory [5], attention [6], sensory and predictive processing [7], and anesthesia [8].

Canonical microcircuits (CMC) model the dynamics of cellular "communities" or subpopulations within a column. Each "neuron" within the model represents the dynamics of a community of neurons with the same morphological phenotype (e.g., pyramidal cell), localized to the same laminar layer, within the same microcolumn, and with a similar functional tuning preference. Neurons within a population have a common functional preference such that stimuli will activate the population to produce a coherent response, a phenomenon verified empirically using electrophysiology techniques [2].

CMC models have evolved over the years to include more complexity, and importantly a tight balance between excitatory and inhibitory interactions [9]. Each distinct microcircuit provides feedforward, feedback, lateral inhibition, and disinhibition. Parallel and serial information processing occurs simultaneously amongst subnetworks, capable of oscillations thought to contribute to higher order cognitive functions and behaviors [10]. Here, we translate the CMC model into a format where it can be solved using the nODE approach [11].

B. Neural ODE Models

Neural Ordinary Differential Equations (Neural ODEs) provide a powerful framework for modeling continuous-time dynamical systems. Unlike traditional neural networks with discrete layers, Neural ODEs define continuous transformations through learnable differential equations. This approach offers several advantages for modeling biological systems: they naturally handle irregular time series, maintain biophysical plausibility through continuous dynamics, and often require fewer parameters than traditional deep learning approaches. Furthermore, Neural ODEs can effectively capture the recurrent, oscillatory nature of cortical processing.

II. METHODS & RESULTS

A. Single Cortical Microcolumn

We implemented the CMC model into neural ODE format incorporating four distinct neural populations: spiny stellate cells, superficial pyramidal cells, inhibitory interneurons, and deep pyramidal cells. Each population is characterized by voltage and conductance states, resulting in an 8-dimensional dynamical system. The model was implemented using a modified sigmoid activation function to mimic neural firing rates, with learnable parameters for extrinsic and intrinsic connectivity strengths and time constants.

Our goal was to then evaluate, optimize and accelerate this model by testing this on a series of classic benchmark AI data sets. We first trained a single CMC “node” on the MNIST dataset using the adjoint sensitivity method for backpropagation through the ODE solver, achieving up to 96.77% test accuracy. Input images were projected into the model’s state space through a learnable transformation, and the final neural states were mapped to classification outputs. Analysis of the model’s behavior revealed distinct trajectories in phase space for different digit classes, suggesting the emergence of digit-specific attractor dynamics, albeit with a loss surface that is not smooth (figure 1).

Canonical Microcircuit (CMC) Neural ODE

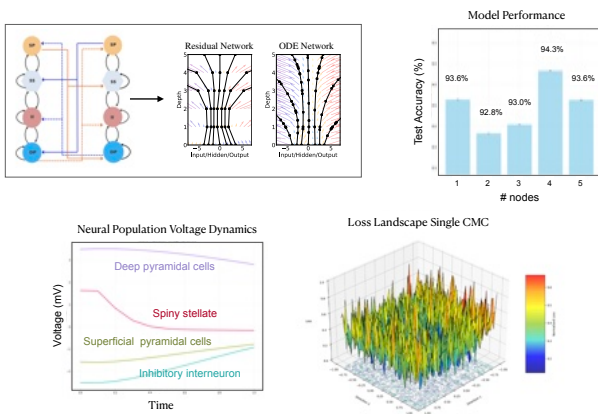


Fig. 1. Top Left) The Canonical microcircuit model was successfully implemented in neural ODE format, [11]. (Top Right) This neural ODE implementation demonstrated competitive performance on the MNIST benchmark data set, and (Bottom left) produced voltage dynamics for neural populations within the cortical micro column. (Bottom Right) Loss landscape for single CMC node.

B. Hierarchical Model

We then extended our neural ODE model to create a hierarchical architecture inspired by the human visual processing stream (V1-V5). The hierarchical model consists of interconnected canonical microcircuit nodes, where each node maintains the core eight-dimensional dynamical system (voltage and conductance states for four neural populations). Inter-regional connectivity was implemented through learnable weight matrices that transform the output states of each region before passing them to the subsequent region. To evaluate the benefits of hierarchical processing, we conducted a systematic comparison of architectures with increasing numbers of regions (1-5) across three standard computer vision datasets: MNIST, CIFAR-10, and CIFAR-100. For each architecture configuration, we performed ten independent training runs to assess consistency and statistical significance. The model integrates input data through an initial projection layer, processes information through the cascaded canonical circuits, and combines the states from all regions through a final readout layer. The loss landscapes for increasing numbers of CMC nodes are shown in Figure 2.

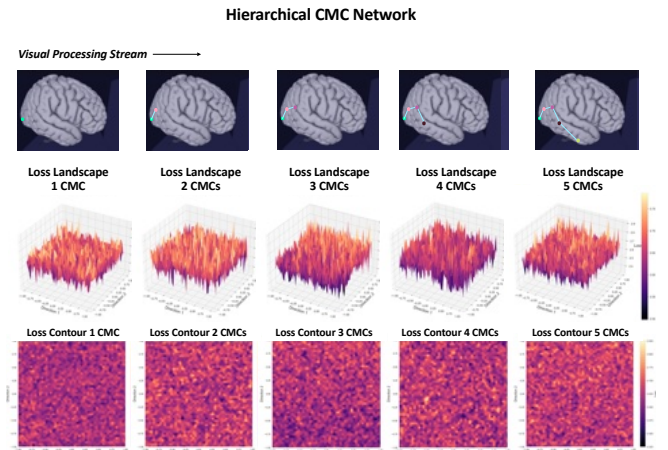


Fig. 2. A Hierarchical CMC model was implemented with 1-5 connected nodes (from left to right), to resemble V1-V5 visual processing stream, showing demonstrating the ability to link CMCs nodes hierarchically, shown with corresponding loss surface and loss contour plots.

C. Recurrent Hierarchical Model

We extended our canonical microcircuit neural ODE to create a recurrent architecture inspired by the bidirectional connectivity patterns observed in the brain’s visual hierarchy. The model incorporates both feedforward and feedback connections between canonical microcircuit nodes, where each node maintains the core eight-dimensional dynamical system capturing voltage and conductance states across four neural populations. Inter-regional connectivity was implemented through learnable weight matrices that enable both bottom-up and top-down state transformations between regions, reflecting the reciprocal nature of cortical processing. We then repeated our systematic comparison of architectures with increasing numbers of regions (1-5) to mimic the layers of processing in the human visual system.

Multiple independent training runs were performed for each configuration to assess robustness and statistical

significance. The model processes information through an initial projection layer, followed by bidirectional interactions between canonical circuits at different hierarchical levels, before integrating states across all regions through a final readout layer. The loss landscape visualization of the recurrent hierarchical CMC demonstrated well-behaved smoother optimization characteristics, with clear convergence basins. Smoother loss surfaces often correlate with better generalization performance and a reduced likelihood of the optimizer getting stuck in a local minimum. Thus, the more biologically plausible recurrent CMC model is likely more robust to parameter perturbations, and will likely perform better on noisy data (Figure 3).

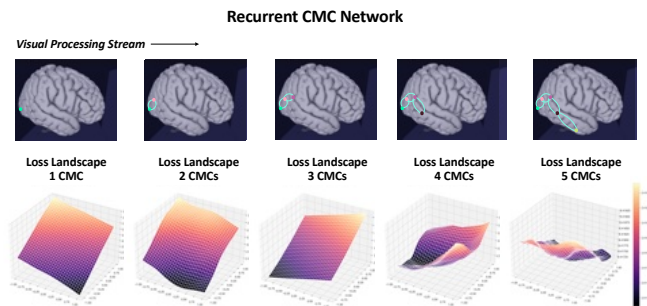


Fig. 3. The Recurrent model adds bi-directional dynamic inter-regional connectivity, which more closely resembles biology and shows significantly smoother loss surfaces, which typically correlate with robust learning and generalization capabilities. Shown with increasing the number of nodes from 1-5 from left to right.

III. DISCUSSION

In this paper, we present a Neural ODE framework for modeling cortical dynamics that bridges biological plausibility with functional capability. Our approach captures the essential dynamics of neural populations while remaining computationally efficient. We demonstrate that even a single CMC node could learn to discriminate hand-written digits with competitively performance. Through systematic evaluation on standard datasets, we show how hierarchical and recurrent extensions of the base model improve performance. Adding recurrence to the model produced smoother loss landscapes, and thus are more likely to generalize to out-of-sample data. Overall, this architecture successfully bridges classical neural models from theoretical neuroscience with modern deep learning approaches. We anticipate that adding more nodes to each functional region (e.g., V1) will provide richer neural dynamics and improved

performance. Future work may involve adding skip connections, and the presence of ephaptic coupling [12],[13] between ODE outputs with equivalent structural connectivity.

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